

Absolutely. Since your goal is to become an **AI Engineer**, let's build something that looks like a real startup product rather than a college project.

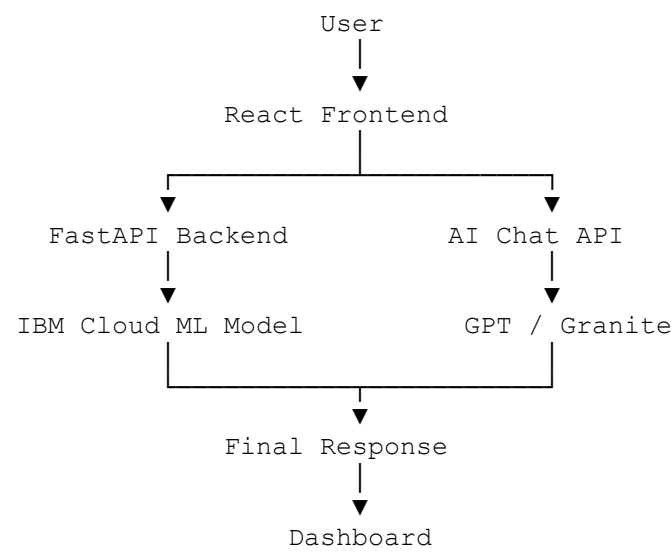
Project Name

BitSense AI — AI-Powered Bitcoin Forecasting Assistant

Tech Stack

- 🚀 Frontend: React + Vite + Tailwind CSS
- ⚡ Backend: FastAPI
- 📦 AI: GPT-5.5 / IBM Granite
- ☁ ML: IBM Cloud AutoAI (your deployed model)
- 🗄 Database: MongoDB Atlas
- 📈 Charts: TradingView + Chart.js
- 📰 News: CoinGecko + CryptoPanic APIs
- 🚀 Deployment: Vercel + Render

Final Architecture



Phase 1 — Backend (Days 1–3)

Folder Structure

```
backend/
├── app/
```

```
|
|
|   api/
|       predict.py
|       chat.py
|       auth.py
|
|   services/
|       ibm_service.py
|       ai_service.py
|       news_service.py
|
|   models/
|
|   schemas/
|
|   database/
|
|   main.py
|
| .env
| requirements.txt
| README.md
```

Backend APIs

Predict

POST /predict

Input

```
{
  "open":40022,
  "high":40246,
  "low":40010,
  "adj_close":40126,
  "volume":22263900160
}
```

Output

```
{
  "prediction":40190.52
}
```

AI Chat

POST /chat

User asks

Should I buy Bitcoin today?

Backend will

Call IBM Model

↓

Prediction

↓

LLM

↓

Explanation

Market Data

GET /market

Returns

- Current Price
 - High
 - Low
 - Volume
-

News

GET /news

Returns

Latest crypto headlines.

Phase 2 — Frontend (Days 4–7)

frontend/

Home

Dashboard

Prediction

AI Chat

History

Settings

Home Page

Current Bitcoin Price

Prediction Card

AI Recommendation

Latest News

Price Chart

Prediction Form

Open

High

Low

Volume

↓

Predict

Result Card

Predicted Close

Confidence

Trend

Bullish

Bearish

Neutral

Phase 3 — AI Agent (Week 2)

User asks

Should I invest today?

Agent

↓

Calls IBM Model

↓

Reads latest news

↓

Reads live price

↓

Explains

Example response

Based on today's market data and recent news, the predicted closing price is around **\$40,190**. Momentum appears slightly bullish, but cryptocurrency prices remain volatile. Consider your own risk tolerance before making investment decisions.

Phase 4 — Live Data

Use

CoinGecko API

Fetch

- Current Price
- 24h High
- 24h Low
- Market Cap

Phase 5 — News

CryptoPanic API

Show

Top Headlines

Positive

Negative

Neutral

Phase 6 — Charts

TradingView widget

OR

Chart.js

Display

Candlestick

Prediction Line

Historical Price

Phase 7 — Database

MongoDB

Store

Prediction History

User Queries

Chat History

Saved Forecasts

Phase 8 — Authentication

Firebase Auth

or

JWT

Phase 9 — Deployment

Backend

Render

Frontend

Vercel

IBM Model

Already deployed ✓

MongoDB

Atlas

Final UI

BitSense AI

Current Price

Prediction

Buy / Hold / Sell

AI Explanation

Chart

News

Ask AI

Prediction History

GitHub Repository Structure

BitSense-AI/

backend/

frontend/

docs/

README.md

LICENSE

Resume Project

BitSense AI – AI-Powered Bitcoin Forecasting Assistant

- Deployed an IBM AutoAI time-series forecasting model as a cloud API.
 - Built a FastAPI backend to integrate the deployed model.
 - Developed a React dashboard with live Bitcoin prices, interactive charts, and prediction history.
 - Integrated an LLM to explain predictions in natural language.
 - Added real-time crypto news and market sentiment analysis.
 - Deployed the full-stack application using Vercel, Render, and IBM Cloud.
-

30-Day Milestones

Week	Goal	Deliverable
1	Backend + IBM API	Working <code>/predict</code> endpoint
2	React dashboard	Users can request predictions
3	AI chat + live market data	Natural-language explanations
4	News, charts, deployment	Public portfolio-ready app

This project demonstrates the complete AI engineering workflow:

- Machine learning deployment
- API integration
- Backend engineering
- Frontend development
- LLM-powered AI agent
- Cloud deployment
- Real-time data integration

It will be a strong centerpiece for your portfolio and a great talking point in interviews because it showcases end-to-end AI application development rather than just model training.